

SmartCounsel: AI-Driven College Admission Predictor for KCET

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Abstract— The Karnataka Common Entrance Test (KCET) is a competitive entrance exam for students studying the engineering stream and seeking admission to various undergraduate courses in the state. Admission to such colleges is finalized based on cutoff ranks, which change annually, and the reservation category of the candidate. One of the biggest challenges that the student is faced with during the KCET counselling process is the uncertainty involved in assuming what colleges and courses they would be eligible for, due to variations in cutoff trends. To address this challenge, the model suggested, SmartCounsel, assists the students by taking their KCET rank and reservation category as input, then predicting a list of probable colleges and courses that they would be likely to be eligible for, based on historical trends. While the initial experimentation involved the use of a Decision Tree for prediction, the final model utilizes a Random Forest Regressor because it has higher accuracy and the capability to handle non-linear data patterns. SmartCounsel assists in minimizing uncertainty during the counselling process and provides students with a data-based solution for college selection.

Keywords— Machine learning (ML), Decision tree model, Random Forest

I. INTRODUCTION

Karnataka Common Entrance Test (KCET), conducted by the Karnataka Examination Authority (KEA), is a highly competitive exam offering admissions to undergraduate degrees in courses such as Engineering, Agriculture, Pharmacy, and Nursing. Thousands of aspiring students compete for a few seats annually, with allotments made on ranking of candidates and reservation categories. However, the dynamic nature of cutoff ranks, based on seat supply, category-wise demand, and institution prestige, makes the admission process complicated and unpredictable for students. One of the major challenges facing the aspirants is how to discover the group of institutions and courses for which they qualify based on their performance and reservation status. Such uncertainty tends to lead to confusion and uninformed judgments during the counselling process. With a view to overcoming this challenge, our paper introduces an AI-powered admission forecast system, SmartCounsel, that leverages historical cutoff data to offer personalized college suggestions. With their KCET rank and reservation category as input, students are given a list of likely colleges and courses

for which they qualify. The model is trained using machine learning algorithms—namely a Random Forest Regressor—to accurately model intricate patterns in the dataset and improve prediction accuracy. The goal of this work is to offer students evidence-based information that aids in making informed decisions during the counselling process. The system removes ambiguity and improves transparency, and thus offers informed and confident choices in choosing higher education institutions.

II. LITERATURE REVIEW AND OVERVIEW

A few research studies have explored machine learning applications in educational counseling and college admission prognosis systems. [1] has explained the use of algorithms like Random Forest, XGBoost, and Multi-layer Perceptron (MLP) to predict college admissions based on historical student data. The authors have mentioned that the machine learning models performed poorly as far as data imbalance was concerned, which led them to have low accuracy. Authors overcame this issue by creating a statistical model that utilized admission patterns and recorded much higher accuracy at 88%. Despite this, the proposed approach uses a dataset obtained only from KCET admission data, thereby ensuring contextual relevance and analytical precision. This article emphasizes the utmost importance of domain-specific datasets in offering predictive models that are specialized for regional entrance examination systems. [2] is a predictive counseling platform designed to assist KCET aspirants in India. The system supports engineering and also offers multilingual access and secure OTP-based login. [3] suggests an artificial intelligence-driven model for predicting graduate admissions based on the examination of past data on universities across different countries. The considered features are Grade Point Average (GPA), Graduate Record Examination (GRE)/Graduate Management Admission Test (GMAT) score, course background, institutional history, English language proficiency (IELTS/TOEFL), and work experience. The authors establish the effectiveness of XGBoost, a machine learning ensemble algorithm, in admission outcome modeling with high accuracy. The research identifies the growing relevance of data-driven decision-making in international education amid the increasing global mobility. Drawing on earlier work in mobile learning and education access, the current research combines machine learning algorithms and

visualization frameworks to improve the predictive accuracy and user engagement of admission advisory systems.

SmartCounsel is designed to guide students through the process of selecting suitable colleges and courses based on KCET rank and reservation category. It uses historical admission data to provide individualized predictions about eligible colleges and courses. The final deployment uses a Random Forest Regressor [4], which was chosen for its improved accuracy and resilience in detecting non-linear relationships. The main goal of SmartCounsel is to reduce uncertainty during the counselling process through an effective, data-driven decision support system. Unlike prior models, the present model includes intuitive ranking bins and visualization layers to enable user-friendly interaction and yield more informative predictive results.

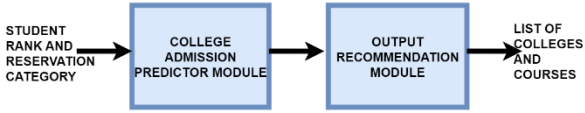


Figure 1: Workflow of SmartCounsel

The SmartCounsel model follows a structured workflow to assist students in making decisions during the KCET counselling process as shown in Figure 1. College admission predictor module shown in Figure 2 begins with collecting historical cutoff data based on college, course, and reservation category. After preprocessing the data—handling missing values, encoding categories, and cleaning inconsistencies—the model explores patterns through EDA. Although initial experiments used decision tree, the final model adopts a Random Forest Regressor for its ability to handle complex, non-linear relationships.

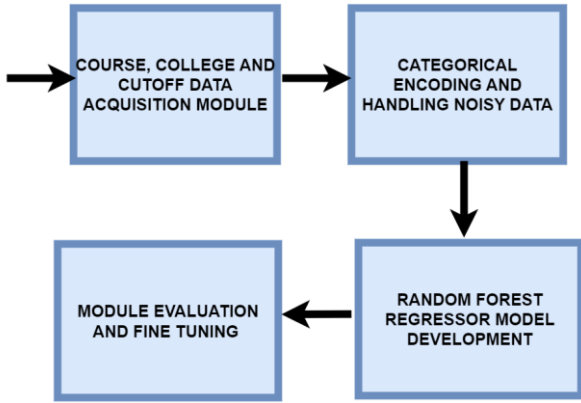


Figure 2: College Admission Predictor Module

The trained model predicts eligibility by comparing the student's KCET rank and category against historical trends, and recommending a list of feasible colleges and courses. A simple interface accepts user inputs and returns colleges recommendations, helping reduce ambiguity and offering a data-driven college selection approach.

III. METHODOLOGY

SmartCounsel utilizes machine learning techniques to predict eligible institutions and courses for KCET aspirants based on historical cutoff data, student rank, and reservation category. The complete methodology consists of the following major phases:

A. Data Acquisition and Preprocessing

The dataset used includes key features such as student rank, course, seat availability and reservation category. The Preprocessing steps involved are Records with missing or invalid cutoff ranks (e.g., zero values) were removed to ensure data integrity, Categorical features such as Institution and Course names were encoded into numeric values using label encoding to facilitate model training, redundant attributes were removed, and essential features contributing to prediction accuracy were retained and correlation analysis was performed to confirm the relevance of selected features. For the purpose of predictive modeling, the minimum cutoff rank observed across the dataset for each course-category combination was selected as the representative threshold. This approach ensures that the model is trained on the most competitive admission benchmarks, thereby enhancing its accuracy in forecasting admission likelihoods under constrained seat availability.

Let the dataset where: $D = \{(r_i, c_i, s_i, q_i, y_i)\}_{i=1}^N$ where r_i denotes the student rank for record i , c_i represents the course or branch, s_i indicates the seat availability for the corresponding course and institution, q_i is the reservation category (e.g., General, SC/ST, OBC), y_i is the cutoff rank for admission in year i . For each unique combination of course c and category q , define the set of cutoff ranks over the years:

$$Y_{c,q} = \{y_i \mid c_i = c, q_i = q\}.$$

The least cut off rank used for prediction is:

$y_{c,q}^* = \min(Y_{c,q})$. This value serves as a benchmark cutoff for predictive modelling, representing the most competitive admission threshold observed across the dataset.

B. Model Selection and Training

Five machine learning models—Decision Tree, Random Forest regressor, Gradient Boosting [5], Support Vector Regression (SVR) [6], and K-Nearest Neighbors (KNN) [7]—were used to predict average college cutoff ranks. Performance was evaluated using RMSE (lower is better) and R^2 score (closer to 1 is better) [8]. Random Forest Regressor the most accurate for predictions and selected for deployment and the same is shown in Figure 3.

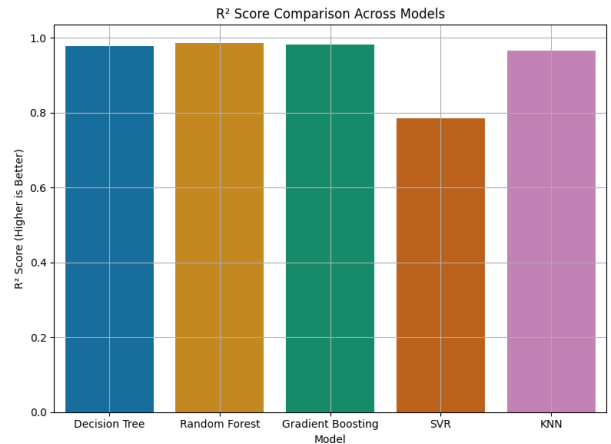


Figure 3: R^2 score Comparison across models

Random Forest achieved the best performance with an RMSE of 8720.42 and an R^2 score of 0.985, indicating high predictive accuracy. Gradient Boosting ranked second, showing competitive results but slightly lower precision. SVR, Decision Tree, and KNN exhibited weaker performance, with higher error rates and lower R^2 scores as shown in Figure 4.

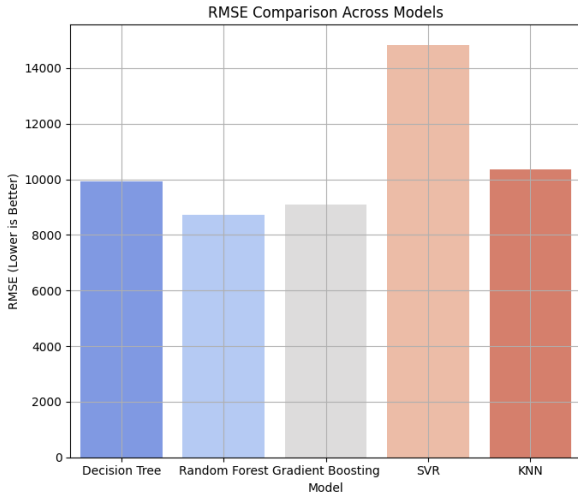


Figure 4: RMSE comparison across models

The dataset D is partitioned into training and testing sets $D_{train} = 80\%$ of D , $D_{test} = 20\%$ of D . The Random Forest model M_{RF} is defined as an ensemble of T decision trees:

$$M_{RF}(x) = 1/T \sum_{t=0}^T h_t(x)$$

Where $h_t(x)$ is the prediction from the t decision tree and $T = n_{estimators} = 100$

The model was evaluated using the following metrics: Root Mean Squared Error (RMSE) and Coefficient of Determination (R^2 Score). The trained model achieved a R^2 score between 0.85 and 0.98, indicating high predictive accuracy, and a low RMSE in the range of 8000–18000.

C. User Interaction and Visualization

Upon entering the KCET Rank and Category, the system predicts the average cutoff rank for each Institution–Course–Category combination. These predictions are sorted and grouped into intuitive rank ranges to aid clarity and comparison. Visualizations such as scatter plots and bar graphs are generated to depict actual versus predicted ranks. This helps candidates assess their admission chances across institutions more effectively. Final recommendations are exportable in structured formats like CSV and Excel for reference during counselling.

IV. RESULTS AND DISCUSSION

The proposed system, **SmartCounsel**, was evaluated on a dataset comprising historical KCET cutoff ranks across various institutions, courses, and reservation categories. The primary objective was to predict eligible college–course combinations for a given KCET rank and category input, thereby assisting students during the counselling phase.

A. Visualization of Predictions

A scatter plot, as illustrated in Figure 5, depicts the relationship between the predicted and actual cutoff ranks. The concentration of data points along the diagonal reference line indicates a high degree of concordance between the model's predictions and the observed values, thereby validating the model's predictive accuracy.

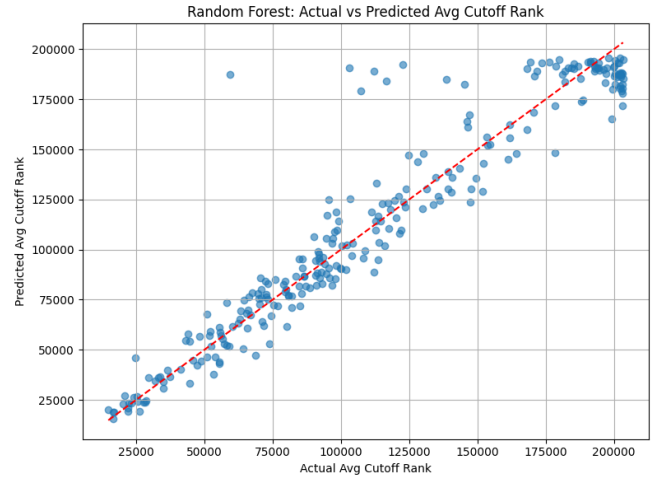


Figure 5: Actual Vs Predicted Average cutoff rank

Figure 6 presents a histogram of prediction errors, revealing a near-normal distribution centered at zero. This symmetric error profile indicates that the model maintains a balanced bias-variance tradeoff, thereby affirming its robustness and reliability in predictive tasks.

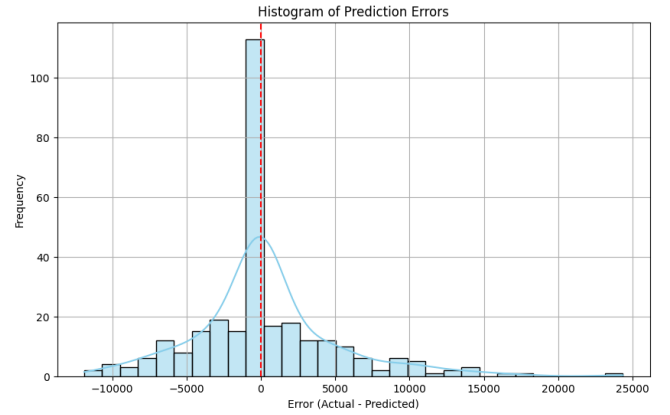


Figure 6: Histogram of prediction errors

B. Category-Wise Analysis

To evaluate model generalizability, predictions were analyzed separately across multiple reservation categories, including General Merit (GM), Scheduled Caste–General (SCG), Scheduled Tribe–General (STG), and Category 2A–General (2AG). The model demonstrated consistent performance across all categories, with only minor variations in Root Mean Squared Error (RMSE) attributable to differences in data volume and distribution. Notably, the GM category, which had the highest data availability, exhibited the most stable and accurate predictions, confirming the model's reliability under well-represented conditions.

Let the reservation categories be denoted as $C = \{GM, SCG, STG, 2AG\}$. For each category c belongs to C Let D_c represent the dataset comprising n_c samples. Empirical results indicate that $RMSE_{GM} < RMSE_c$ for most c belongs to $C \setminus \{GM\}$ due to higher n_{GM} .

C. Rank-Bin Based Prediction Output

To enhance interpretability and decision-making, the user interface categorized predicted college–course combinations into discrete **rank-based bins**:

$B1$	$Top\ 10,000$
$B2$	$10,001 \leq Rank \leq 25,000$
$B3$	$25,001 \leq Rank \leq 50,000$
$B4$	$50,001 \leq Rank \leq 100,000$
$B5$	$100,001 \leq Rank \leq 200,000$
$B6$	$Rank > 200,000$

Let r_u denote the rank of a given user. The binning function $B(r_u)$ maps each user to a subset of predicted options:

$$B(r_u) = \{(c_j, s_j) | r_j \text{ belongs to } B_k\}$$

Where c_j is the predicted college, s_j is the associated course, r_j is the predicted cutoff rank for (c_j, s_j) and B_k is the bin corresponding to r_u .

D. Comparative Merits and Model Limitations

Compared to conventional static lists, the suggested system offers a dynamic, data-driven, and personalized approach to predicting college admissions. The addition of real-time filtering and sorting features significantly reduces ambiguity during the KCET counselling process by allowing students to work interactively with their available options. Additionally, the addition of in-built visualization tools like the scatter plot and bar graph, along with CSV exportability, significantly enhances usability, thus allowing students to save and share their results with peers, mentors, or counsellors for improved guidance. Though its applied utility, the model has some limitations. Its predictive power is effectively dependent on the availability, completeness, and reliability of past cutoff data. Any errors, missing values, or outliers in the data can adversely affect model performance and generate suboptimal recommendations. Additionally, the implementation does not account for real-time changes to the seat matrix, changes in institutional quotas, or variations in student preferences, all of which affect actual admission results.

E. Graphical User Interface

The KCET prediction application is powered by a web-based graphical user interface which has been optimized to provide ease of use and performance boost. The frontend is created using Flask, with Jinja2 templates being used to provide dynamic rendering of HTML content. The users are required to input their KCET rank through a numeric field and their category through a dropdown menu. Upon submission, the data is sent to the backend through HTTP requests. The backend, created with FastAPI, processes the input and invokes a machine learning model to predict eligible colleges. The predicted results are returned as a structured dataset and then displayed as a formatted HTML table. The table has the institute name, course, city, and cutoff rank for each eligible college. A "Download CSV" button is also available to enable users to download the filtered list for offline use. CSS is used to style the interface, thus providing responsiveness and visual simplicity. The entire application is served through the Uvicorn ASGI server, which provides efficient and low-latency performance.

F. Test Case Analysis

Category based Prediction Case 1:

In this test scenario, the user selects the reservation category "3AG" and inputs a KCET rank of 13000 into the prediction

form illustrated in Figure 7. Upon submission, the system processes the input and generates a ranked list of Institution–Course–Category tuples with predicted average cutoff ranks. This test validates the system's ability to handle category-specific inputs and produce accurate, personalized recommendations aligned with the candidate's profile.

SmartCounsel: AI-Driven College Admission Predictor for KCET

Select Category:

Enter Your Rank:

Figure 7: Sample prediction form – 3AG category

The system generates a structured table listing all eligible colleges based on the user's KCET rank and selected category. Each row in the table corresponds to a valid Institution–Course–Category tuple.

Download CSV			
Institute	Course	City	Cutoff
Siddaganga Institute of Technology Tumkur	Artificial Intelligence And Data Science	Tumakuru	13425
K.L.S. Gogte Institute of Technology Belgaum	Computers Science And Engineering	Belgaum	13528
Sir M.Visweswaraya Institute of Technology Hunasemaranahalli, Bangalore,	Computers Science And Engineering	Bengaluru	14017
Dr. Ambedkar Institute Of Technology Jnanabharathi Campus, Bangalore	Computers Science And Engineering	Bengaluru	14926
Sir M.Visweswaraya Institute of Technology Hunasemaranahalli, Bangalore,	Artificial Intelligence and Machine Learning	Bengaluru	16996
Nitte Meenakshi Institute of Technology Bangalore	Artificial Intelligence And Data Science	Bengaluru	14136
Sri Dharmasthala Manjunatheswara College of Engineering Dhanwad	Artificial Intelligence and Machine Learning	Dhanwad	20686
Sri Dharmasthala Manjunatheswara Collene	Computers Science And		

Figure 8: Sample list of eligible Institution–Course for 3AG Category

As illustrated in Figure 8, a functional "Download CSV" button is provided, enabling users to export the filtered list into a CSV file for offline reference and use during the counselling process. This feature enhances usability by allowing candidates to retain and share personalized recommendations in a standardized format.

General merit Category prediction Case 2:

In this test case, the user selects the reservation category "GM" (General Merit) and enters a KCET rank of 13000 into the prediction form as shown in Figure 9.

SmartCounsel: AI-Driven College Admission Predictor for KCET

Select Category:

Enter Your Rank:

Figure 9: Sample prediction form – GM category

Upon submission, the system processes the input and retrieves a list of eligible Institution–Course–Category

combinations based on historical cutoff data as shown in Figure 10.

Download CSV			
Institute	Course	City	Cutoff
Sir M. Visveswaraya Institute of Technology Hunasemaranahalli, Bangalore.	Computers Science And Engineering	Bengaluru	13629
Dr. Ambedkar Institute Of Technology Jnanabharathi Campus, Bangalore	Computers Science And Engineering	Bengaluru	14803
Sir M. Visveswaraya Institute of Technology Hunasemaranahalli, Bangalore.	Artificial Intelligence and Machine Learning	Bengaluru	16922
Nitte Meenakshi Institute of Technology Bangalore	Artificial Intelligence And Data Science	Bengaluru	13937
Sri Dharmasthala Manjunatheswara College of Engineering Dharwad	Artificial Intelligence and Machine Learning	Dharwad	15056
C M R Institute of Technology Bangalore	Computers Science And Engineering	Bengaluru	13866
Basaveshwara Engineering College Bagalkot	Computers Science And Engineering	Bagalkot	16243
R. V. College of Engineering Bangalore	Chemical Engineering	Bengaluru	14949

Figure 10: Sample list of eligible Institution–Course for GM Category

A **Download CSV** button allows users to export this filtered list into a .csv file, enabling offline access and reference during the counselling process as shown in Figure 11.

1	InstituteName	CourseName	City	GM
2	sveswaraya Institute of Technology Hunasemaranahalli	Computers Science And Engineerin	Bengaluru	13629
3	bedkar Institute Of Technology Jnanabharathi Campus	Computers Science And Engineerin	Bengaluru	14803
4	sveswaraya Institute of Technology Hunasemaranahalli	Artificial Intelligence and Machine Lear	Bengaluru	16922
5	Nitte Meenakshi Institute of Technology Bangalore	Artificial Intelligence And Data Scien	Bengaluru	13937
6	armasthala Manjunatheswara College of Engineering	Artificial Intelligence and Machine Lear	Dharwad	15056
7	C M R Institute of Technology Bangalore	Computers Science And Engineerin	Bengaluru	13866
8	Basaveshwara Engineering College Bagalkot	Computers Science And Engineerin	Bagalkot	16243
9	R. V. College of Engineering Bangalore	Chemical Engineering	Bengaluru	14949
10	bedkar Institute Of Technology Jnanabharathi Campus	Artificial Intelligence and Machine Lear	Bengaluru	18685
11	armasthala Manjunatheswara College of Engineering	Information Science and Engineerin	Dharwad	14026
12	Siddaganga Institute of Technology Tumkur	ronics and Communication Enginee	Tumakuru	13211
13	bedkar Institute Of Technology Jnanabharathi Campus	Information Science and Engineerin	Bengaluru	16568
14	S K S J T Institute of Engineering, Bangalore	Computers Science And Engineerin	Bengaluru	23163
15	R. V. College of Engineering Bangalore	Civil Engineering	Bengaluru	18609

Figure 11 : Sample .csv file to enable offline access

V. CONCLUSION

This paper presents an artificial intelligence-based college admission prediction system, carefully designed to streamline the Karnataka Common Entrance Test (KCET) counselling process. An average cutoff rank prediction is made by the system for every Institution–Course–Category triple, as a function of a candidate's KCET rank and reservation category, based on a Random Forest Regressor trained on past cutoff data. While the predictions are currently limited to average cutoff values and do not take year-to-year differences in college-specific quotas into account, the model has shown an impressive level of predictive accuracy. For improved transparency and to enable decision-making, the system provides personalized forecasts, rank-bin

categorization, and visualization features in the form of scatter plots and bar graphs. In comparison with the conventional static document or manual search methods, this dynamic method reduces uncertainty to a large degree and enables students to make extremely sound study decisions.

Subsequent additions to enhance the scope and precision of the system are:

Simulating student preferences and seat assignment to provide a closer examination of admissions, Including institution-level information, i.e., course popularity patterns and seat matrix, to enhance the level of granularity in predictions, Gathering voluntary user feedback during counseling to allow real-time model adjustment, Embedding the model in a web-based interactive application to enhance accessibility and scalability at the Karnataka level, and Including marks from boards, seat matrix changes, and institution preferences as features to allow enhanced forecasting capabilities.

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